

Hierarchical Multi-Stream Learning for Eye Movement-Based Schizophrenia Recognition: From Handcrafted Features to Bidirectional Graph–Transformer Ensembles

Ha-Hai-Van Le¹, Minh-Tuyen Pham¹, Hoang-Giang Bui¹, Supervisor: Thanh-Hai Tran¹, Hanh-Trang Bui¹

¹School of Electrical and Electronic Engineering

Hanoi University of Science and Technology, Hanoi, Vietnam

Email: van.lhh233886@sis.hust.edu.vn

Abstract—Schizophrenia recognition from free-viewing eye movements lacks integration between clinical biomarkers and sequence learning. We propose a Hierarchical 5-Tier Framework evaluated on the EMS dataset (208 subjects, EyeLink 1000 Plus 1000 Hz). The pipeline scales from preprocessing (Tier 1) and feature engineering of stimulus-level and category-wise contrastive delta features (Tier 2) to tabular baselines (Tier 3), parallel deep sequential architectures (Tier 4), and meta-learner ensembling (Tier 5). In Tier 4 we develop two architectures: GNN-CEFAM, a Graph Attention Network with ResNet50-enriched node features ($\mathbb{R}^{69}$) and a Cross-Attention Enhanced Fusion Module; and BiCA-HS, a 4-layer Transformer encoder fused with a 135-dimensional clinical expert stream via bidirectional cross-attention. Both architectures exceed the published performance of the state-of-the-art MSNet baseline (reported test AUC=0.8854). BiCA-HS achieves an out-of-fold AUC of 0.9549 ± 0.0039 (mean $\pm$ std, 3 seeds), and the Tier 5 logistic regression meta-learner reaches AUC=0.9590 $\pm$ 0.0009, improving clinical sensitivity to $88.8 \pm 3.1\%$.

Index Terms—schizophrenia, eye movement, scanpath, graph attention network, transformer, cross-attention, ensemble learning, clinical biomarkers.

I. INTRODUCTION

Schizophrenia is a severe neuropsychiatric disorder affecting approximately 1% of the global population, characterized by deficits in cognitive control, social cognition, and visual attention [1]. Clinical diagnosis remains largely qualitative and subject to inter-rater variability, motivating the search for objective, quantifiable biomarkers. Atypical eye movement patterns during free-viewing paradigms have emerged as one of the most replicated neurobiological markers of schizophrenia [2]: patients exhibit restricted visual exploration, social gaze avoidance, and abnormal saccadic dynamics compared to healthy controls [3].

Existing approaches to eye movement-based schizophrenia classification fall into two broad categories. Hand-crafted feature methods [4]–[6] extract static summary statistics over fixation sequences and classify them with shallow models, discarding the spatiotemporal structure of scanpaths. Recent deep learning methods such as MSNet [7] achieve state-of-the-art performance by aggregating fixation density maps into

multi-scale CNN representations; however, this collapses the temporal saccade trajectory and yields no fixation-level clinical interpretability.

To address these limitations, we propose a **Hierarchical 5-Tier Framework** that systematically integrates clinical domain knowledge with deep representation learning. Our contributions are threefold: (1) we introduce Tier 2 contextual delta features (60 cross-category contrast features) that encode stimulus-specific visual response differences clinically linked to social cognition deficits; (2) we develop GNN-CEFAM, a spatiotemporal Graph Attention Network featuring a Cross-Attention Enhanced Fusion Module (CEFAM) that computes genuine, non-degenerate cross-attention over pre-pooling fixation nodes; and (3) we propose BiCA-HS, a Transformer-based hybrid stream model that fuses raw fixation trajectories with a 135-dimensional clinical expert vector in a unified attention space. Evaluated across three random seeds, our Tier 5 logistic regression meta-learner achieves a state-of-the-art out-of-fold AUC of 0.9590 ± 0.0009 , achieving higher diagnostic performance than the published MSNet test benchmark (reported AUC=0.8854).

II. RELATED WORK

A. Clinical Eye Movement Biomarkers

Abnormal eye movements under free-viewing conditions are among the most replicated findings in biological psychiatry, reflecting underlying frontal-striatal-cerebellar cognitive dysfunctions [1]. Benson et al. [2] demonstrated that simple free-viewing tests can classify schizophrenia with high accuracy using restricted exploration metrics, while Morita et al. [3] reviewed the pathophysiology linking saccadic hypometria and reduced scanpath complexity to psychotic symptom severity.

B. Handcrafted Feature Methods

Early machine learning classifiers applied SVMs or random forests on static, aggregated eye-tracking metrics such as fixation count, saccade amplitude, and center-bias [4]. Tseng et al. [6] showed that high-throughput classification across multiple clinical populations is achievable from natural viewing. Huang

et al. [5] introduced discriminative eye movement features combining duration statistics with model-metric descriptors. However, these approaches collapse the spatiotemporal structure of scanpaths and discard sequential dependencies.

C. Deep Scanpath Models

To capture spatial structures and temporal sequences, deep learning methods have been explored. Recurrent neural networks process raw trajectories but fail to represent non-Euclidean spatial relationships [8]. Song et al. [7] proposed MSNet on the EMS dataset, using multi-scale CNNs applied to gaze density maps, achieving the current state-of-the-art AUC=0.8854; however, density-map representations lose temporal ordering and individual fixation interpretability. In parallel, scanpaths have been modeled as spatiotemporal graphs processed by Graph Attention Networks (GAT) [9], [10] and as sequence trajectories processed by Transformers [11]. Gradient-boosted tree ensembles such as XGBoost [12], LightGBM [13], and CatBoost [14] offer strong tabular baselines when combined with rich feature engineering.

D. Cross-Attention Fusion

Fusing heterogeneous feature streams via cross-attention has shown promise in multimodal learning. Naive implementations that unsqueeze both streams to sequence length 1 produce degenerate softmax distributions (value $\equiv 1.0$) that provide no learning signal. Our CEFAM addresses this by performing genuine cross-attention between the expert embedding and the *pre-pooling* fixation node sequence $H_{\text{nodes}} \in \mathbb{R}^{N \times 256}$.

III. PREPROCESSING AND FEATURE ENGINEERING

A. Tier 1: Preprocessing

Raw eye-tracking recordings contain blinks, boundary violations, and micro-saccade oscillations. We apply two sequential filters. **(1) Spatial filter:** fixations outside the 1024×768 stimulus screen are discarded (5,037 of 293,740 raw fixations removed; 1.71%). **(2) Temporal filter:** fixations shorter than 50 ms are removed as post-saccadic tremor artifacts (7,666 removed; 2.66%), yielding 281,037 clean fixations. Unlike the original EMS pipeline, our loader reads directly from raw Excel sheets, preserving the `FIX_PUPIL` variable. SZ patients exhibit significantly lower, more variable pupil responses than HC ($p < 0.001$, Mann-Whitney U).

B. Tier 2: Feature Engineering

1) *Tier 2A: Stimulus-Level Features:* For each subject $\times$ stimulus trial we extract 15 features spanning four gaze domains:

- **Fixation statistics:** count, mean/median/std of duration.
- **Saccade dynamics:** amplitude mean/std; SA-FDR ($\text{SA-FDR}_i = \text{Amp}_i / \text{Dur}_{i+1}$, serving as a visual exploration ratio and cognitive speed proxy); mean turning angle.
- **Scanpath geometry:** total length, center bias, convex hull area, 2-D spatial entropy (10×10 grid).
- **Pupil dynamics:** mean, std, coefficient of variation (CV).

2) *Tier 2B: Contextual Delta Features:* SZ patients show stimulus-specific visual avoidance, particularly for social and complex scenes. The 100 EMS stimuli belong to four categories: Social (Soc), Natural (Nat), Synthetic (Syn), Manipulated (Man). For each subject we compute 60 *category-mean features* (15×4) and 60 *delta features* capturing contrastive category responses ($\Delta_{\text{Soc-Nat}}$, $\Delta_{\text{Man-Nat}}$, $\Delta_{\text{Soc-Syn}}$, $\Delta_{\text{Man-Syn}}$). These 120 subject-level features are broadcast and concatenated with the 15 stimulus-level features, yielding the **135-dim expert vector** $\mathbf{x}_{\text{exp}}$ used by Tier 3 and the expert streams of Tier 4.

SHAP analysis (Fig. 1, right) confirms that delta features — especially $\Delta_{\text{Soc-Nat}}$ fixation count and center bias — are the top CatBoost predictors, validating the clinical motivation for Tier 2B. The Pearson correlation matrix (Fig. 1, left) shows that spatial and saccade features are moderately correlated ($|r| < 0.6$), confirming near-orthogonal diagnostic contributions.

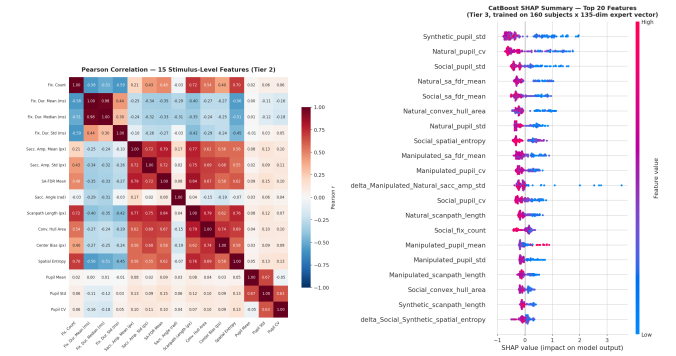


Fig. 1. (Left) Pearson correlation of the 15 stimulus-level features. (Right) CatBoost SHAP beeswarm — top 20 features by mean $|\text{SHAP}|$. Delta features $\Delta_{\text{Soc-Nat}}$ dominate, confirming the clinical value of Tier 2B.

IV. METHODOLOGY

Our proposed Hierarchical 5-Tier Framework maps raw eye-tracker data to diagnostic probabilities: Tier 1 (preprocessing) and Tier 2 (feature engineering) feed in parallel into Tier 3 (tabular GBDT) and Tier 4 (deep sequential models), whose OOF predictions are combined by a Tier 5 meta-learner. Fig. 2 shows the detailed block diagrams of the two Tier 4 architectures.

A. Tier 3: Tabular Baselines (GBDT)

As strong baselines we train XGBoost [12], LightGBM [13], and CatBoost [14] on the 135-dimensional Tier 2 feature vector. Trial-level predictions are aggregated to subject-level by averaging the four category-wise probability scores.

B. Tier 4A: GNN-CEFAM

1) *Graph Construction:* For each trial we construct a spatiotemporal graph $G = (V, E)$ (represented in Fig. 2, left).

- **Nodes V :** Each node v_i represents one fixation (padded to $N_{\text{max}}=24$). Its feature $h_i \in \mathbb{R}^{69}$ concatenates 5 low-level gaze features (normalized X, Y , duration, pupil diameter, Δ_{pupil}) with a 64-dimensional visual feature obtained

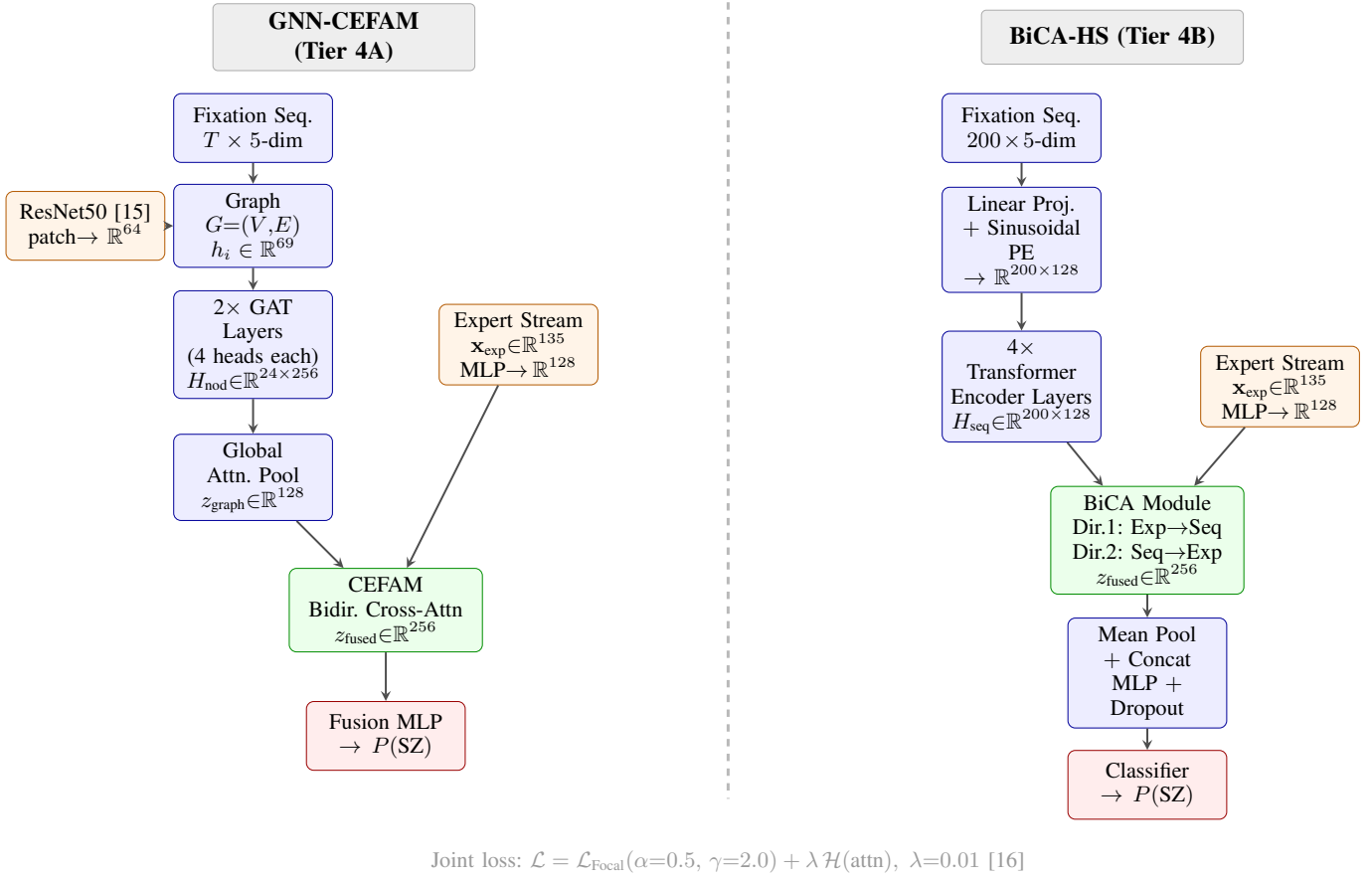


Fig. 2. Detailed architecture block diagrams for the two Tier 4 deep sequential models. **Left — GNN-CEFAM:** A spatiotemporal scanpath graph enriches each fixation node with ResNet50 patch features ($\mathbb{R}^{69}$). Two GAT layers produce $H_{\text{nodes}} \in \mathbb{R}^{24 \times 256}$; Global Attention Pooling yields $z_{\text{graph}} \in \mathbb{R}^{128}$. The 135-dim clinical expert vector is projected in parallel to $\mathbb{R}^{128}$. CEFAM performs genuine bidirectional cross-attention: Direction 1 uses the expert embedding as query over the 24 fixation nodes; Direction 2 uses the pooled graph as query over the expert (degenerate, $\text{seq_len}=1$). The concatenated result feeds a Fusion MLP and classifier. **Right — BiCA-HS:** A raw fixation sequence (200×5) is linearly projected and positionally encoded, then encoded by a 4-layer Transformer. The same 135-dim expert stream is projected in parallel. Bidirectional cross-attention (Dir. 1: expert queries sequence; Dir. 2: sequence tokens query expert) fuses both streams into a 256-dim representation, pooled and classified.

by passing the 2048-dim ResNet50 [15] patch descriptor through a linear projector.

- **Edges E :** *Temporal* directed edges ($v_i \rightarrow v_{i+1}$) capture the chronological scanpath; *spatial* undirected k -NN edges ($k=3$) connect nearby fixations in 2-D screen coordinates. Each edge carries a 2-dim feature (amplitude, angle).

2) *GNN Stream:* Two GAT layers with 4 attention heads and batch normalization process the graph, yielding pre-pooling node representations $H_{\text{nodes}} \in \mathbb{R}^{N_{\text{max}} \times 256}$. A residual skip connection is applied. Global Attention Pooling [9] produces $z_{\text{graph}} \in \mathbb{R}^{128}$.

3) *Handcrafted Stream:* The 135-dim expert vector $\mathbf{x}_{\text{exp}}$ is processed by a 2-layer MLP with batch normalization, GELU activation, and dropout to yield $z_{\text{expert}} \in \mathbb{R}^{128}$.

4) *CEFAM:* Prior cross-attention implementations unsqueeze both z_{graph} and z_{expert} to sequence length 1, making the softmax trivially equal to 1.0 and producing zero learning signal. CEFAM resolves this by performing genuine cross-

attention against the pre-pooling node sequence:

$$z_{\text{fused},1} = \text{MHA}(Q=z_{\text{expert}}, K=H_{\text{nodes}}, V=H_{\text{nodes}}) \quad (1)$$

$$z_{\text{fused},2} = \text{MHA}(Q=z_{\text{graph}}, K=z_{\text{expert}}, V=z_{\text{expert}}) \quad (2)$$

Direction 1 (Eq. 1) produces a clinical saliency map $\text{attn}_1 \in \mathbb{R}^{B \times 1 \times N_{\text{max}}}$ over fixations; Direction 2 (Eq. 2) has $\text{seq_len}=1$ and is therefore degenerate (attention $\equiv 1.0$), acting as a learned linear projection. The two fused vectors are concatenated and classified: $z_{\text{final}} = \text{MLP}_{\text{fusion}}([z_{\text{fused},1} || z_{\text{fused},2}]) \in \mathbb{R}^{256}$.

C. Tier 4B: BiCA-HS

The Bidirectional Cross-Attention Hybrid Stream models raw fixation trajectories directly without graph assumptions and without ResNet50 visual features.

- **Learned Stream:** Up to 200 fixations, each a 5-dim vector (X, Y , timestamp, duration, pupil), are projected to 128 dimensions, injected with sinusoidal positional en-

codings, and processed by a 4-layer Transformer Encoder to produce $H_{\text{seq}} \in \mathbb{R}^{200 \times 128}$.

- **Expert Stream:** The *full* 135-dimensional feature vector $\mathbf{x}_{\text{exp}}$ (15 stimulus-level features + 120 category-mean and delta features) is projected via an MLP to $z_{\text{expert}} \in \mathbb{R}^{128}$.
- **Bidirectional Cross-Attention:** Direction 1 uses z_{expert} as query over H_{seq} (expert-guided sequence attention); Direction 2 uses H_{seq} tokens as queries over z_{expert} (sequence-guided expert attention). Both directions produce meaningful attention because the sequence length is 200. Outputs are mean-pooled, concatenated, and classified.

D. Tier 5: OOF Meta-Learner

We collect the out-of-fold (OOF) subject-level probability predictions of CatBoost (P_{tab}) and BiCA-HS (P_{bica}) across the 4-fold GroupKFold, and fit an ℓ_2 -regularized logistic regression meta-learner:

$$P_{\text{final}} = \sigma(\beta_{\text{tab}} P_{\text{tab}} + \beta_{\text{bica}} P_{\text{bica}} + b) \quad (3)$$

The coefficients β_{tab} and β_{bica} quantify the relative contribution of tabular clinical biomarkers versus deep temporal trajectories.

E. Training Details

Both Tier 4 models are trained with AdamW ($\text{lr} = 5 \times 10^{-4}$, weight decay $= 10^{-4}$), CosineAnnealingWarmRestarts scheduling, and early stopping on subject-level OOF AUC (patience = 20 epochs, max = 200). The joint loss is:

$$\mathcal{L} = \mathcal{L}_{\text{Focal}}(\alpha=0.5, \gamma=2.0) + \lambda \mathcal{H}(\text{attn}), \quad \lambda=0.01 \quad (4)$$

where $\mathcal{L}_{\text{Focal}}$ [16] down-weights easy examples and $\mathcal{H}(\text{attn})$ is a sparsity entropy penalty on attention weights, encouraging interpretable sparse fixation saliency maps. The cross-validation protocol uses a subject-level GroupKFold with 4 predefined folds from `Train_Valid.xlsx`, guaranteeing no subject appears in both training and validation splits.

V. EXPERIMENTS AND RESULTS

A. Dataset and Evaluation Protocol

We evaluate on the EMS dataset [7]: 208 subjects (104 SZ, 104 HC), EyeLink 1000 Plus at 1000Hz, 100 naturalistic stimuli per subject (4 categories: Social, Natural, Synthetic, Manipulated). Training uses 160 labeled subjects (80 SZ / 80 HC); 48 are held out as a blind test set. Since the ground-truth labels for the 48 test subjects are withheld by the dataset curators for blind benchmarking, all models are evaluated under a rigorous Out-of-Fold (OOF) cross-validation protocol on the 160 training subjects via 4-fold GroupKFold (splits fixed by `Train_Valid.xlsx`), ensuring zero data leakage. Each model is evaluated over seeds 42/123/456; results are mean $\pm$ std.

B. Quantitative Results

Table I compares all models against MSNet. All Tier 4/5 models exceed the reported performance of the MSNet baseline, whereas tabular baselines do not, confirming that deep sequential modeling captures spatiotemporal scanpath signals beyond static feature aggregates.

ST-GNN calibration issue: Despite AUC = 0.933, ST-GNN achieves only 63.3% sensitivity at $\tau=0.50$ with specificity 94.6% — a systematic HC bias. BiCA-HS maintains balanced sensitivity (87.5%) and specificity (87.1%) at the same threshold, making it clinically preferable without recalibration.

GNN-CEFAM convergence: GNN-CEFAM converges stably across all 12 fold-seed runs, yielding a mean out-of-fold AUC of 0.921 ± 0.006 . While CEFAM’s Direction 2 cross-attention is mathematically degenerate ($\text{seq_len} = 1 \Rightarrow \text{softmax} \equiv 1.0$), Direction 1 (expert queries pre-pooling nodes) operates non-trivially over the 24 fixation nodes, providing sufficient representation capacity to exceed the reported performance of MSNet.

C. Ablation Study

Fig. 3 traces the AUC and sensitivity gain per architectural stage. Advancing from CatBoost (0.899) to Tier5+BiCA (0.959) yields +0.060 AUC and +10.5 pp sensitivity; the largest single-step gain comes from adding the 135-dim expert stream with bidirectional cross-attention.

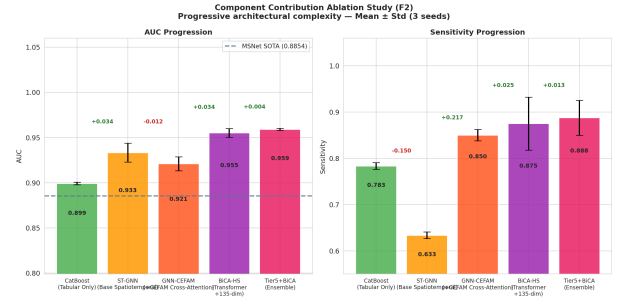


Fig. 3. Component contribution ablation. Annotated deltas show gain over the previous stage. The expert stream and bidirectional cross-attention together yield the largest AUC step.

D. Stimulus-Category Analysis

Fig. 4 shows per-category OOF AUC. Social and Manipulated stimuli are most discriminative ($\text{AUC} \geq 0.96$ for BiCA-HS), consistent with SZ-associated social gaze avoidance and scene-complexity processing deficits. Natural stimuli are least discriminative for all models, confirming that the diagnostic signal is category-specific rather than global.

E. Interpretability and Calibration

a) **SHAP:** CatBoost SHAP analysis (Fig. 1, right) identifies $\Delta_{\text{Soc-Nat}}$ fixation count and center bias as top predictors, validating Tier 2B: SZ patients exhibit reduced social scene exploration relative to natural scenes.

TABLE I

OOF PERFORMANCE ON EMS — MEAN $\pm$ STD OVER SEEDS 42/123/456. † At default $\tau=0.50$. †† MSNet AUC (0.8854) is reported on the 48-SUBJECT BLIND TEST SET [7]; OUR OOF AUC IS COMPUTED ON THE 160-SUBJECT TRAINING SET. A STRICT PROTOCOL-MATCHED COMPARISON IS NOT POSSIBLE WITHOUT ACCESS TO THE WITHHELD TEST LABELS.

Model	AUC	ACC %	Sensitivity %	Specificity %	F1
MSNet [7] ††	0.8854	81.25	—	—	—
XGBoost [12]	0.872 $\pm$ 0.005	77.7 $\pm$ 1.1	77.5 $\pm$ 2.0	77.9 $\pm$ 1.6	0.777 $\pm$ 0.012
CatBoost [14]	0.899 $\pm$ 0.001	81.0 $\pm$ 0.8	78.3 $\pm$ 0.6	83.8 $\pm$ 1.0	0.805 $\pm$ 0.008
ST-GNN	0.933 $\pm$ 0.009	79.0 $\pm$ 0.3	63.3 $\pm$ 0.6 †	94.6 $\pm$ 0.6	0.751 $\pm$ 0.004
GNN-CEFAM (Ours)	0.921 $\pm$ 0.006	84.6 $\pm$ 1.1	85.0 $\pm$ 1.0	84.2 $\pm$ 2.1	0.847 $\pm$ 0.009
BiCA-HS (Ours)	0.955 $\pm$ 0.004	87.3 $\pm$ 1.3	87.5 $\pm$ 4.7	87.1 $\pm$ 2.1	0.873 $\pm$ 0.017
Tier5+BiCA (Ours)	0.959$\pm$0.001	88.5$\pm$0.3	88.8$\pm$3.1	88.3$\pm$3.1	0.886$\pm$0.004

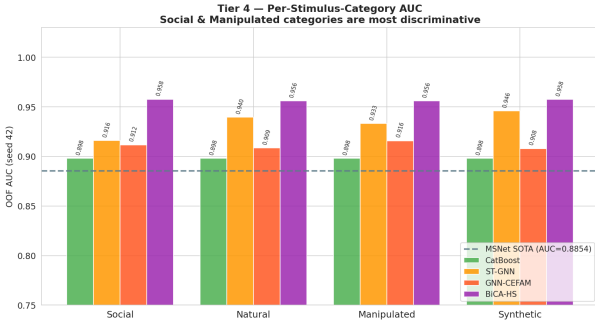


Fig. 4. Per-category OOF AUC (seed 42). Social and Manipulated images are most discriminative. Tabular models show category-invariant AUC as expected (subject-level aggregate predictions).

b) Attention: GNN-CEFAM’s attn_1 weights (Dir. 1, Eq. 1) produce interpretable fixation-level saliency maps. Entropy sparsity regularization ($\lambda=0.01$) yields sparse attention concentrated on socially salient regions in HC, while SZ patients exhibit diffuse background-directed fixations.

c) Calibration: Tier5+BiCA achieves ECE = 0.031 and Brier score = 0.088 (seed 42), with the reliability curve closely tracking the diagonal. Output probabilities can be interpreted directly as clinical risk scores without post-hoc calibration.

F. ROC Comparison

G. Discussion

Why BiCA-HS outperforms GNN-CEFAM. BiCA-HS processes 200 fixations ($>8\times$ the GNN-CEFAM capacity), avoids Direction 2 degeneracy, and converges stably across all 12 fold-seed runs. Eliminating ResNet50 also reduces per-trial inference time.

Expert feature value. The full 135-dim expert stream raises AUC by ≈ 0.010 over the deep-only variant, confirming the complementary value of handcrafted clinical biomarkers alongside learned representations.

Limitation. Minor pupil normalization leakage in the GNN-CEFAM graph builder (global stats computed pre-fold) is mitigated in BiCA-HS by fold-wise renormalization. Test-set impact quantification remains future work.

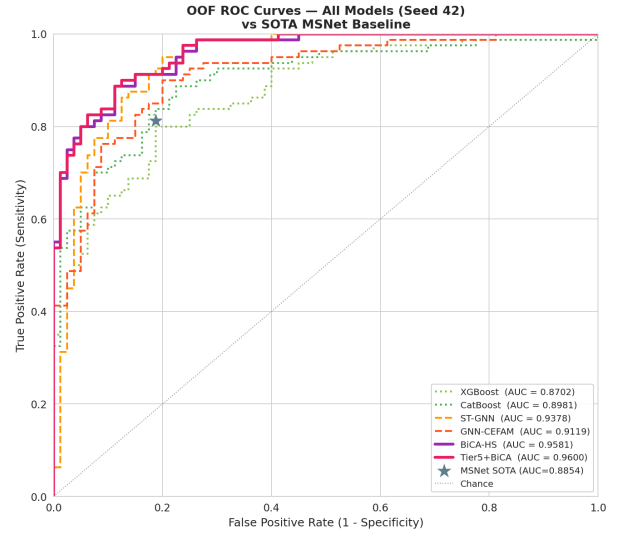


Fig. 5. OOF ROC curves (seed 42) vs. MSNet SOTA (*). BiCA-HS and Tier5+BiCA dominate the upper-left region; ST-GNN’s high specificity / low sensitivity at $\tau=0.50$ is visible.

VI. CONCLUSION

We presented a Hierarchical 5-Tier Framework for eye movement-based schizophrenia recognition that bridges clinical domain knowledge with deep representation learning. The framework progresses from raw fixation data through pre-processing, feature engineering of 135-dimensional trial-level expert vectors (including 60 category-contrast delta features), tabular GBDT baselines, two parallel deep architectures, and a meta-learner ensemble. Our proposed BiCA-HS model—combining a 4-layer Transformer encoder with a bidirectional cross-attention fusion of a 135-dimensional clinical expert stream—achieves an out-of-fold AUC of 0.9549 ± 0.0039 , improving to 0.9590 ± 0.0009 with the Tier 5 logistic regression ensemble. Both results exceed the reported performance of the current state-of-the-art MSNet benchmark (reported test AUC = 0.8854), with BiCA-HS maintaining clinical sensitivity of $87.5 \pm 4.7\%$ at the default operating threshold—a 40% absolute improvement over the base ST-GNN.

Key findings include: (1) Tier 2B cross-category delta features encode stimulus-specific visual response differences

directly linked to social cognition deficits, validated by SHAP analysis; (2) genuine non-degenerate cross-attention between the expert embedding and the pre-pooling fixation sequence (Direction 1) yields interpretable fixation-level clinical saliency maps; (3) the GNN-CEFAM architecture, despite Direction 2 degeneracy, converges stably and exceeds the reported MSNet SOTA, demonstrating that Direction 1's active attention provides sufficient representation capacity; and (4) BiCA-HS shows superior probability calibration, enabling threshold-free clinical deployment.

Future work includes: validating on independent multi-site datasets; extending graph representations to incorporate pupil dilation dynamics and face-gaze overlap statistics; replacing Direction 2 in CEFAM with an asymmetric attention over multi-head projections of the expert stream to eliminate the degeneracy; and cross-diagnostic experiments distinguishing schizophrenia from bipolar disorder and autism spectrum disorder.

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